

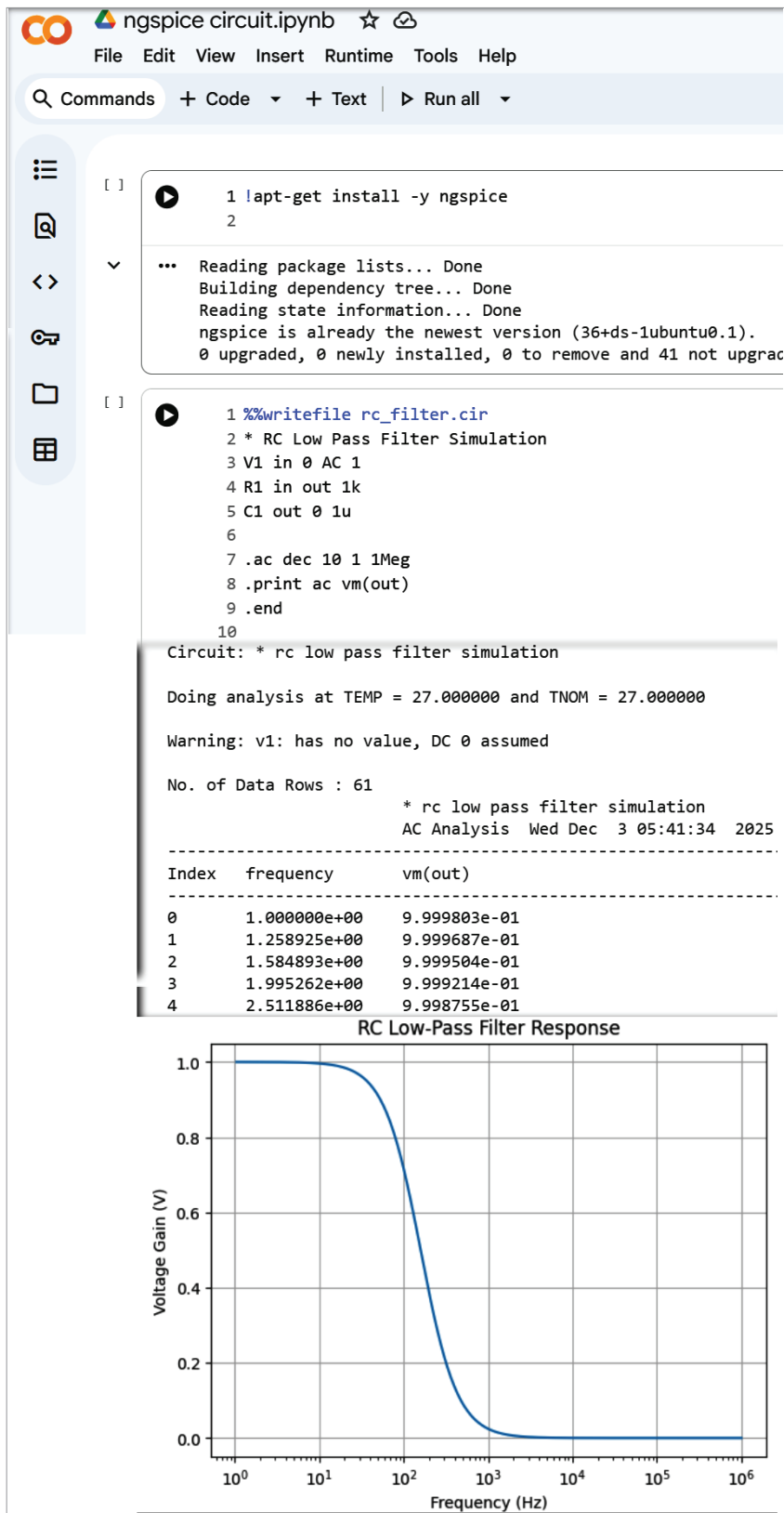


Figure 5: Generating a circuit on Google Colab using ngspice code created by ChatGPT

```

w, h = 2.0, 1.0
# Draw rectangle
dr += els.Line().right().at((x, y)).length(w)
dr += els.Line().down().length(h)
dr += els.Line().left().length(w)
dr += els.Line().up().length(h)
# Add text label inside the rectangle
dr += els.Label().at((x + w/2, y - h/2)).label('PMU', halign='center', valign='center')
dr.pop()
# Continue bus to Smart Grid
dr += els.Line().right().length(1.0)
dr += els.Dot(open=True)
dr += els.Line().right().length(0.8).label('To Smart Grid', loc='right')
# Load connected at grid side
dr += els.Line().down().length(0.9)
dr += els.Resistor().label('Local Load', loc='right')
dr += els.Line().down().length(0.2)
dr += els.Ground()
dr.draw()

```

Generating circuits using ngspice

Ngspice (<https://ngspice.sourceforge.io/>) is a prominent library for the generation of a range of electrical and electronic circuits for carrying out research. Ngspice scripts can be generated using AI chatbots and executed on Google Colab (Figure 5).

Simulation of circuits using pandapower

Another free and open source library called pandapower (<https://www.pandapower.org/>) can be used for power system modelling and optimisation for better automation and performance. Just like ngspice and SchemDraw, it can be used to program any kind of complex circuit or model for research.

These script-based Python libraries can be used freely for modelling and simulating circuits